

An Alternative Analysis of the Morphosyntactic and Semantic Structure of Japanese Nominal Adjectives — *NI*-adjectives¹

Shinya Makino

itakanro@g.ecc.u-tokyo.ac.jp

Abstract:

The present paper reanalyses the morphosyntactic and semantic structure of Japanese nominal adjectives (NAs), also referred to as *na*-adjectives. **The core of NAs lies in the NA stem in *-ni***, as in *yukai-ni* 'to be pleasant' and *suki-ni* 'to like'. These NA stems are lexical predicates that require arguments — the argument structure [theme] inherent in *yukai-ni* and [experiencer, theme] in *suki-ni*. However, outside small clauses functioning as complements of cognition or perception verbs, as in *watasi=wa [sore=o yukai-ni] omot-ta / kanzi-ta* 'I found / felt [it pleasant]', such NA stems cannot function as predicates in their bare *-ni* form.

Consequently, an NA stem forms a nominal-adjective phrase accompanied by PRO, as in [*PRO yukai-ni*], which serves as the complement of the adjunctive conjunction *-te*, yielding a ConjP [*PRO yukai-ni-te*]. This ConjP is then adjoined to the VP headed by the one-argument existential verb *ar-* within a topic-marked, tensed construction, as in [*sore_i=wa t_i ar-O*] 'it exists'. Only in this configuration can the NA stem function as the predicate of an independent sentence, as in [*sore_i=wa [PRO_i yukai-ni-te] t_i ar-O*], which surfaces as *sore=wa yukai-d-a* [*sore_i=wa [PRO_i yukai-d] t_i a-O*] '(lit.) it, being pleasant, exists', schematically representable as *EXIST(it) ∧ PLEASANT(it)* — whose default negation pattern is partial, namely *EXIST(it) ∧ ¬PLEASANT(it)*, realised as *sore=wa yukai-de=wa nai* '(lit.) it, not being pleasant, exists'.

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1. Introduction

Japanese **nominal adjectives** (NAs), also referred to as **na-adjectives** due to their attributive forms ending in *-na* — the so-called *rentaikei* in Japanese — as in *sizuka-na heya* 'a quiet room', have often been analysed as "so-called NA stems + the so-called copula =*da*" in parallel with "NPs + the so-called copula =*da*", as in *sizuka=da* 'it is quiet' and *gakusei=da* 'it is a student'. This analysis, however, seems to have limitations, in that it cannot account for the fact that **only in the inflexion of NAs do forms ending in -ni exist**, as in *sizuka-ni* 'quiet; quietly', whereas in the inflexion of the so-called copula =*da* following NPs there are none, since forms such as *gakusei=ni* 'in/at/to/from a/the student' are postpositional phrases (PPs) headed by the postposition =*ni*. The present paper argues that **it is precisely this -ni which constitutes the defining constituent of NAs**, and undertakes an analysis of their morphosyntactic and semantic structure.

2. The identification of the NA stem, NA radical and NA thematic suffix

Traditionally, in Japanese **verbal adjectives** (VAs) — also referred to as adjectives, canonical adjectives or **i-adjectives**, owing to their attributive forms ending in *-i*, as in *taka-i tō* 'a tall tower' — inflected forms, such as the conjunctive form *taka-ku* from *takai* 'to be high/tall' (so-called *renyōkei*), have been analysed as consisting of the so-called stem *taka-* and the so-called ending *-ku*. In Makino (2025¹: 6), by contrast, given **the multiple morphological, syntactic and semantic parallels with verbal stems** such as *tab-e* from *taberu* 'to eat', it is assumed that *taka-ku* as, a whole, functions as **the stem of a verbal adjective** — henceforth, **VA stem**.

The conjunctive form *sizuka-ni* of the NA *sizukada* 'to be quiet/silent', constituted by the so-called nominal-adjective stem *sizuka-* followed by the so-called ending *-ni*, can occur in the same syntactic-semantic environments as the VA stem *taka-ku* — analysed in Makino (2025¹: 7–8) as consisting of the **VA radical** *taka-* followed by the **VA thematic suffix**, or **stem-forming suffix**, *-ku* — as shown in (1) – (6), where '-' and '=' indicate 'morpheme boundary in general' and 'enclisis', respectively:

(1) *Kare=wa sizuka-ni mat-ta*. 'He danced quietly.'

(2) *Kare=wa taka-ku mat-ta*. 'He danced high.'

- (3) sizuka-ni nar-u / sizuka-ni su-ru 'to become/make quiet '
- (4) taka-ku nar-u / taka-ku su-ru 'to become/make high/tall'
- (5) Watasi=wa sore=o yukai-ni [0past] omot-u. 'I find it pleasant.'
- (6) Watasi=wa sore=o tanosi-ku [0past] omot-u. 'I find it enjoyable.'
- (7) Watasi=wa sore=wa/ga yukai-d-a=to [-past] omot-u. 'I think that it is pleasant.'
- (8) Watasi=wa sore=wa/ga tanosi-i=to [-past] omot-u. 'I think that it is enjoyable.'
- (9) Watasi=wa sore=wa/ga yukai-d-at-ta=to [+past] omot-u. 'I think that it was pleasant.'
- (10) Watasi=wa sore=wa/ga tanosi-k-at-ta=to [+past] omot-u. 'I think that it was enjoyable.'

In (1) and (2), both *sizuka-ni* and *taka-ku* function as **adverbial adjuncts of the verbal predicate** *mat-* — an allomorph of the stem *maw-* from the consonant-stem verb *ma(w)-u* 'to dance gracefully or ceremonially' — and qualify the manner of dancing, whereas, in (3) and (4), both constitute **complements of the verbal predicates** *nar-* and *su-* from the state-change verbs *nar-u* 'to become' and *su-ru* 'to make', denoting the results of certain state changes.

In (5) and (6), on the other hand, both *yukai-ni* and *tanosi-ku* function as **atemporal predicates without temporal-modal endings** in the small clauses *sore=o yukai-ni* 'it pleasant' and *sore=o tanosi-ku* 'it enjoyable', respectively, which serve as complements of *omo-*, an allomorph of the verbal predicate *omow-* 'to think/find'. (Note that the inflexional endings *-ru* $\approx$ *-u*, specified as [-past], and *-ta*, specified as [+past], following *maw-*, *nar-*, *su-* and *omow-* in (1) – (10) are tense suffixes and function as the heads of tense phrases at the sentential level, syntactically independently of such verbal stems.) The verbal predicate *omow-* can take as its complements not only small clauses, as in (5) and (6), but also tensed complementiser phrases (CPs) headed by *=to* 'that'. In (7) and (8), *omow-* takes as its complements the [-past] CPs *sore=wa/ga yukai-da=to* 'that it is pleasant' and *sore=wa/ga tanosi-i=to* 'that it is enjoyable', whereas, in (9) and (10), it takes as its complements the [+past] CPs *sore=wa/ga yukai-dat-ta=to* 'that it was pleasant' and *sore=wa/ga tanosi-k-at-ta=to* 'that it was enjoyable'.

The relationship between the demonstrative *sore* and the VA inflected forms *tanosi-ku*, *tanosi-i*, *tanosi-k-at-ta* in (6), (8), (10) is, in all cases, one between a theme and its state. If *tanosi-i* and *tanosi-k-at-ta* are stative predicates, then *tanosi-ku* must likewise be a stative predicate.

Similarly, the relationship between *sore* and NA inflected forms *yukai-ni*, *yukai-d-a*, *yukai-d-at-ta* in (5), (7), (9) is in all cases the same one between a theme and its state. If *yukai-d-a* and *yukai-d-at-ta* are stative predicates, then *yukai-ni* must likewise be a stative predicate. It is therefore evident that ***yukai-ni*, as well as *tanosi-ku*, functions not only as an adverbial adjunct or a complement to verbal predicates, but also as a predicate in its own right.**

It is demonstrated by the accusative marking of its complement *sore* with the case particle =*o*, as in *sore=o*, that *yukai-ni*, like *tanosi-ku*, is an atemporal form without a tense suffix. In Japanese, tense suffixes are assumed to function as nominative-case assigners, and nominative case is assigned to the specifier of the tense phrase, as stated in Watanabe (2009: 121, translated from Japanese): "In Japanese ... nominative case is assigned to the NP occupying the specifier of TP." Since *tanosi-i* and *yukai-d-a* are specified as [–past], and *tanosi-k-at-ta* and *yukai-d-at-ta* are specified as [+past], by virtue of their paradigmatic oppositions, each can be considered to contain a tense suffix specified as [–past] or [+past], respectively.

The complement *sore* 'it' — which is not assigned any case within the phrases headed by the NA and the VA in (7) – (10), since neither NAs nor VAs are case assigners in Japanese — moves to the specifier position of the phrase headed by these tense suffixes, where it receives nominative case and surfaces as *sore=ga* with the nominative-case particle =*ga*; whereas, if the complement *sore* moves instead to the specifier position of the topic phrase headed by the phonologically-null topicalisation suffix, it surfaces as *sore=wa*. By contrast, *yukai-ni* and *tanosi-ku* contain no tense suffix and thus the complement *sore* remains within the phrase headed by those forms, without movement to the tense phrase. Consequently, *sore* is assigned accusative case by the transitive predicate *omow-* and surfaces as *sore=o*. From this, it follows that the **NA inflected forms ending in *-ni* can function as atemporal predicates**, and that **it seems reductive to refer to them merely as adverbial forms**, as has traditionally been done.

It is thus evident that in these examples both *sizuka-ni* and *taka-ku* — as well as *yukai-ni* and *tanosi-ku* — occur in the same phrase-structural positions and fulfil the same semantic functions. If *taka-ku* as a whole functions as a VA stem, as argued in Makino (2025¹: 6), then it can be assumed that *sizuka-ni* as a whole functions in parallel as a nominal-adjective stem — hereafter abbreviated as an NA stem. Just as, in the VA stem *taka-ku*, the constituents *taka-* and *-ku* function as VA radical and VA thematic suffix, respectively, it can be assumed that, in the NA stem *sizuka-ni*, the so-called nominal-adjective stem *sizuka-* in fact constitutes a nominal-

adjective radical — henceforth, NA radical — which encodes the lexical meaning specific to the nominal adjective, and that the so-called ending *-ni* functions as the NA thematic suffix, or VA stem-forming suffix, and licenses the NA radical as a syntactic constituent of category NA at the sentential level. Thus, the NA radical and the NA thematic suffix are the two immediate constituents of the NA stem as indicated in (11):

(11)

	Stem	
	Radical	Thematic suffix
VA	<i>taka-</i>	<i>-ku</i>
NA	<i>sizuka-</i>	<i>-ni</i>

Just as the VA radicals *taka-* and *tanosi-* alone, lacking the VA thematic suffix *-ku*, cannot function as constituents of sentences, so too the NA radicals *sizuka-* and *yukai-* alone, lacking the NA thematic suffix *-ni*, cannot function as constituents of sentences, as shown by the ungrammatical examples in (12) – (17), which correspond to the grammatical ones in (1) – (6):

- (12) **Kare=wa sizuka mat-ta*. 'He danced quietly.' (cf. *sizuka-ni mat-ta*)
- (13) **Kare=wa taka mat-ta*. 'He danced high.' (cf. *taka-ku mat-ta*)
- (14) **sizuka nar-u* / **sizuka su-ru* 'to become/make quiet/silent' (cf. *sizuka-ni nar-u* / *su-ru*)
- (15) **taka nar-u* / **taka su-ru* 'to become/make high/tall' (cf. *taka-ku nar-u* / *su-ru*)
- (16) **Watasi=wa sore=o yukai omo-u*. 'I find it pleasant.' (cf. *yukai-ni omo-u*)
- (17) **Watasi=wa sore=o tanosi omo-u*. 'I find it enjoyable.' (cf. *tanosi-ku omo-u*)

Formally, the NA thematic suffix *-ni*, as in *sizuka-ni*, exhibits the same phonological representation /ni/ as the locative postposition — or, case particle — *=ni*, as in *Tōkyō=ni* 'in/at/to Tokyo', and in fact the former seems to have originated from the latter. Furthermore, the suffix sequence *-ni-te* and the postposition–suffix sequence *=ni-te* suffered the same historical change — namely, *ni-te* → *de*, as in *sizuka-ni-te* and *miyako=ni-te* 'in the capital' → *sizuka-de* and *miyako=de* — probably due to the deletion of *-i-* and the postnasal voicing of *-t-* due to *-n-*: *-ni-te* → *-n-te* → *-n-de* → *-de*. On the other hand, there is a **major functional difference** between the NA stem ending in the NA thematic suffix *-ni* and the PP headed by the postposition *=ni* with an NP complement: the former can function as an adverbial adjunct of predicates,

whereas the latter cannot function as such, but rather as an argument of the predicate as shown in (18) – (20):

(18) *Kare=wa sizuka-ni arui-ta/mat-ta.* 'He walked/danced quietly.'

(19) **Kare=wa yūzin-ni arui-ta/mat-ta.* '*He walked/danced to a friend.'

(20) *Kare=wa yūzin-ni hon=o kasi-ta.* 'He lent a book to a friend.'

If in (20) the PP *yūzin-ni* were replaced by the NA *sizuka-ni*, then the latter would likely not be recognised as such, but rather as the PP *Sizuka=ni* 'to (a woman named) Sizuka', as in (21). Moreover, the NA *sizuka-ni* functioning as an adverbial adjunct and the PP *Sizuka=ni* functioning as an argument can co-occur within a single sentence, as shown in (22):

(21) *Kare=wa Sizuka=ni hon=o kasi-ta.* 'He lent a book to Sizuka.'

(22) *Kare=wa Sizuka=ni sizuka-ni hon=o watasi-ta.* 'He handed a book to Sizuka quietly.'

Therefore, it seems plausible to distinguish functionally the NA thematic suffix *-ni* from the postposition *=ni*: the former, as the **head of stems**, merges with NA radicals — e.g. *sizuka-* 'quietness', *odayaka-* 'calmness', *kantan-* 'easiness' — to form stems of the NA category — e.g. *sizuka-ni* 'quiet', *odayaka-ni* 'calm', *kantan-ni* 'easy'². By contrast, the latter, as the **head of syntactic phrases**, merges with NPs — e.g. *yūzin* 'a friend', *taisetuna yūzin* 'a dear friend', *mukasi=kara=no yūzin* 'a friend from long ago' — to form PPs — e.g. *yūzin=ni* 'in/to/from a friend', *taisetuna yūzin=ni* 'in/to/from a dear friend', *mukasi=kara=no yūzin=ni* 'in/to/from a friend from long ago'.

² With respect to the adverbial use of VAPs and NAPs, it is provisionally assumed that a phonologically zero AdvP head merges with VAPs and NAPs, such as *takaku-ku* and *sizuka-ni*, to form AdvPs, which function as adjuncts of predicates. Semantically, however, the adverbial use of NAPs is not unnatural. If *sizuka-* in *sizuka-ni* is interpreted as 'quietness' and *-ni* as 'in', the expression as a whole can be understood as 'in quietness', that is, 'in a state or situation of quietness', thereby allowing an eventuality to be located within it. At the same time, the fact that *sizuka-ni* can be modified by degree adverbs such as *hizyōni*, *totemo*, *kiwamete* and *hanahada*, as in *hizyōni / totemo / kiwamete / hanahada sizuka-ni* 'very / extremely quietly', makes it difficult to analyse *sizuka-* straightforwardly as an NP and *-ni* as a postposition.

Although morphosyntactically distinct, however, the NA thematic suffix *-ni* seems to share essentially the same semantics with the locative postposition *=ni*: in the sequence "NA radical + *-ni*" it carries the meaning of 'in/at'. For example, the form *siawase*, which can function both as an NA radical and as an NP, forms an NA stem when merged with the NA thematic suffix *-ni*, but forms a PP when merged with the postposition *=ni*, as indicated in (21) and (22):

(23) *Kare=wa kazoku=to siawase-ni kurasite=iru*. 'He is living happily with his family.'

(24) *Kono miti=wa siawase-ni tuduite=iru*. 'This road leads to happiness.'

The bare form *siawase* 'happiness' refers only to a certain mental state, without specifying whether, in a given situation, that state denotes the theme of motion, as in *koko=ni siawase=ga yatte=ku-ru* 'happiness comes here', the object of pursuit, as in *watasi=wa siawase=o oimotome-ru* 'I seek happiness', the goal of a path, as in *kono=miti=wa siawase-ni tuduite=iru* 'this road leads to happiness' in (22), or bears some other semantic roles. Once, however, the NA thematic suffix *-ni* is added to *siawase*, as in (21), the form *siawase-ni* comes to mean not simply 'happiness' but 'in/at happiness'. In other words, ***-ni* 'in/at' converts the NA radical *sizuka-* — which refers to a conceptual place 'quietness' — into a conceptual location 'in/at quietness', thereby making it possible for the theme denoted by a NP to be located at that place, and only at this stage does *siawase*, when merged with *-ni*, function predicatively in the sentence, that is, as a predicative stem**, more specifically as an NA stem, as shown in (25) and (26), where the NA radical *siawase-*, when merged with the NA thematic suffix *-ni*, functions as an atemporal predicate, whereas *siawase* without *-ni* cannot function as such:

(25) *Kanozyo=wa ima=no seikatu=o siawase-ni kanzi-ta*. 'She felt her present life happy.'

(26) **Kanozyo=wa ima=no seikatu=o siawase kanzi-ta*. 'idem.'

It can be concluded from these observations that the **NA thematic suffix *-ni* functions morphosyntactically differently from the formally identical postposition *=ni***, yet **shares the same semantics of 'in/at' with this locative postposition *=ni***, and functions as an **indispensable morphological and semantic constituent for the construction of the NA stem**. Thus, it seems **more appropriate to call NAs *ni*-adjectives** rather than *na*-adjectives, after their defining constituent *-ni*.

3. The sequence of the NA thematic suffix *-ni* and the endings *-te*, *-temo*, *-tatte*

The major difference between the NA stem *sizuka-ni* and the VA stem *taka-ku* is that, whereas both can function as atemporal predicates within small clauses, the former cannot occur alone at the end of non-sentence-final coordinated clauses, as shown in (27) and (28):

(27) *Kanozyo=wa se=ga taka-ku, sosite utukusii/utukusikatta*. 'She is/was tall and beautiful.'

(28) **Kare=wa sizuka-ni, sosite odayakada/odayakadatta*. 'He is/was silent and calm.'

But the inflected form *sizuka-de* 'to be quiet/silent and', corresponding semantically to the VA inflected form *taka-ku-te* 'to be high/tall and' — constituted by the VA stem *taka-ku* followed by the conjunctive suffix *-te* — is able to occur in this position, as indicated in (29) and (30):

(29) *Kanozyo=wa se=ga taka-ku-te, utukusii/utukusikatta*. 'She is/was tall and beautiful.'

(30) *Kare=wa sizuka-de, odayakada/odayakadatta*. 'He is/was silent and calm.'

From the proportional relation *taka-ku* : *taka-ku-te* = *sizuka-ni* : *sizuka-de*, it can be hypothesised that *-de* is a **surface form of the underlying sequence *-ni-te***. Indeed, in ConjPs one can observe stylistic variation between the postposition–conjunction amalgam *=de* 'in, at' and the postposition–conjunction sequence *=ni-te* 'idem.', as in *Tōkyō=de* ≈ *Tōkyō=ni-te* 'in/at Tokyo', the latter being the more formal and literal form. Furthermore, the assumed underlying form *sizuka-ni-te* is historically attested as an actual classical form, which constitutes the etymon of the modern *sizuka-de*. It does not seem implausible, therefore, to assume that the surface form *sizuka-de* derives synchronically from the underlying *sizuka-ni-te*, on the condition that in contemporary language the surface realisation of */-ni-te/* as *[-de]* is obligatory in NA stems. Building on this interpretation, the forms *sizuka-demo* and *sizuka-datte* can be analysed in the same way as *sizuka-de*, as in (31):

(31)	Stem		Ending	Surface form
	Radical	Thematic suffix		
	<i>sizuka-</i>	<i>-ni</i>		
	<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	
	<i>sizuka-</i>	<i>-ni</i>	<i>-temo</i>	
NA	<i>sizuka-</i>	<i>-ni</i>	<i>-tatte</i>	<i>sizuka-datte</i>

VA	<i>taka-</i>	<i>-ku</i>		<i>taka-ku</i>
	<i>taka-</i>	<i>-ku</i>	<i>-te</i>	<i>taka-ku-te</i>
	<i>taka-</i>	<i>-ku</i>	<i>-temo</i>	<i>taka-ku-temo</i>
	<i>taka-</i>	<i>-ku</i>	<i>-tatte</i>	<i>taka-ku-tatte</i>

The problem with this analysis is that the shaded derivation *sizuka-ni-tatte* → *sizuka-datte* is neither diachronically attested nor consistent with our native-speaker intuition. In the present study, nevertheless, for the reasons set out in Section 5, this analysis is maintained.

4. The default negation pattern and the syntactic structure of NA constructions

Between the so-called copular construction in (32) and the nominal-adjectival one in (33):

(32) *Sore=wa gakkō=de ar-u.* 'It is a school.'

(33) *Sore=wa yukai-de ar-u.* 'It is pleasant.'

there is a shared property: the default negation pattern in both cases is one of partial negation, with the scope of negation marked by the particle *=wa*, as shown in (34) and (35):

(34) *Sore=wa gakkō=de=wa (*ara-)**nai**.* 'It is not a school.'

(35) *Sore=wa yukai-de=wa (*ara-)**nai**.* 'It is not pleasant.'

(Note that in (34) and (35), as well as in (36) – (39) below, the stem of the existential verb *aru* is represented as *(*ara-)* in *(*ara-)**nai***, since in Contemporary Standard Japanese the negative form of this verb is not the theoretically expected **ara-nai*, but is obligatorily substituted by the suppletive adjective *nai*.) By contrast, the total negation in (36) and (37):

(36) *Sore=wa gakkō=de (*ara-)**nai**.* 'It is not a school.'

(37) *Sore=wa yukai-de (*ara-)**nai**.* 'It is not pleasant.'

is perceived as infelicitous. In more colloquial language, the constructions in (38) and (39), with the contracted form *dya* derived from the sequence *de=wa* are very often used instead of those

in (34) and (35):

(38) *Sore=wa gakkō=dya (*ara-)nai.* 'It is not a school.'

(39) *Sore=wa yukai-dya (*ara-)nai.* 'It is not pleasant.'

According to Makino (2025²: 4-10), the so-called copular construction in (32) can be syntactically represented as [*sore_i=wa [gakkō=de] t_i ar-u-Top⁰∅*] or, more underlyingly, as in (40):

(40) [*sore_i=wa [gakkō=ni-te] t_i ar-ru-Top⁰∅*]

which consists of the one-argument obligatory construction [*sore_i=wa t_i ar-ru-Top⁰∅*] 'it exists' and the adjunctive construction [*gakkō=ni-te*] 'with (its existence) located in the conceptual domain of "school"' or, more practically, 'and it is as a school', which surfaces as [*gakkō=de*]. The category Topic — abbreviated as Top in (40) — is "a kind of functional category" that "can be considered to embed the IP into discourse and context" (Hasegawa 1999: 91, translated from Japanese), and is assumed in this paper to be realised by the phonologically null morpheme Top⁰∅ -0. The element that moves to the specifier of TopP is assigned the particle =*wa* and functions to indicate the topic of the sentence.

Semantically, this construction is both existence-asserting and existence-domain-specifying. The VP [*sore ar-*] in the syntactically obligatory construction [*sore_i=wa t_i ar-ru-Top⁰∅*] asserts its existence. On the other hand, the underlying postposition =*ni* in the syntactically adjunctive ConjP [*gakkō=ni-te*] behaves as if it were a two-argument predicate, and locates its existence denoted by the VP [*sore ar-*] within the conceptual domain of 'school' referred to by the adjunctive NP [*gakkō*].

Schematically, this construction can be represented as $\text{EXIST}(\text{sore}) \wedge \text{IN}(\text{rEXIST}(\text{sore}), \text{gakkō})$, where the function *r* in $\text{rEXIST}(\text{sore})$ — abbreviated from 'reified' — maps the proposition $\text{EXIST}(\text{sore})$ 'it exists' to the corresponding reified eventuality $\text{rEXIST}(\text{sore})$, that is, an eventive entity representing 'its existence'. The morpheme sequence =*ni-te ar-* as a whole correspond to corresponds to the schematic expression $\text{EXIST}(\text{ }) \wedge \text{IN}(\text{rEXIST}(\text{ }), \text{ })$, witse the predicate-like postposition =*ni*, the adjunctive conjunction -*te* and the existential-verb stem *ar-* represent $\text{IN}(\text{r}(\text{ }), \text{ })$, $\wedge$ and $\text{EXIST}(\text{ })$, respectively.

The default negation pattern of the affirmative construction *sore=wa gakkō=de aru* in (32) is not the total negation *sore=wa gakkō=de (*ara-)nai* in (36), but rather the partial negation *sore=wa gakkō=de=wa (*ara-)nai*. The negative *-nai* does not negate, therefore, the entire construction *sore=wa gakkō=de aru*, but instead targets only the adjunct *gakkō=de*. Schematically, the interpretation is not $\neg(\text{EXIST}(\text{sore}) \wedge \text{IN}(\text{rEXIST}(\text{sore}), \text{gakkō}))$, but rather $\text{EXIST}(\text{sore}) \wedge \neg\text{IN}(\text{rEXIST}(\text{sore}), \text{gakkō})$. Semantically, the sentence continues to assert the existence of *sore* since *sore* exists, while negating only the claim that *sore*'s existence is located in the conceptual domain of *gakkō*. This derives from the adjunctive nature of *gakkō=ni*, which constitutes an additional proposition.

Since the nominal-adjectival construction *sore=wa yukai-de aru* in (33) also takes partial negation — namely, *sore=wa yukai-de=wa (*ara-)nai* in (35) — as its default negation pattern, just like the so-called copular construction *sore=wa gakkō=de aru* in (32), it can likewise be assumed to consist of the existence-asserting, one-argument obligatory construction [*sore_i=wa t_i ar-u-Top⁰Ø*] 'it exists', combined with the state-specifying adjunctive construction [*yukai-ni-te*] 'and (it) is pleasant', which surfaces as [*yukai-de*].

However, there is a syntactic and semantic difference between the PP *gakkō=ni* contained in the underlying form *gakkō=ni-te* of *gakkō=de* in *sore=wa gakkō=de aru* in (32) and the NA stem *yukai-ni* contained in the underlying form *yukai-ni-te* of *yukai=de* in *sore=wa yukai-de aru* in (33). While the NA stem *yukai-ni*, much like the VA stem *tanosi-ku* 'to be enjoyable', can take an argument and function as the predicate of a small clause serving as the complement of cognition or perception verbs such as *omou* 'to think' or *kanziru* 'to feel', the PP *gakkō=ni* cannot, as demonstrated in (41) = (5), (42) = (6) and (43):

(41) *Watasi=wa [sore=o yukai-ni] omot-ta / kanzi-ta*. 'I found/felt [it pleasant].'

(42) *Watasi=wa [sore=o tanosi-ku] omot-ta / kanzi-ta*. 'I found/felt [it enjoyable].'

(43) **Watasi=wa [sore=o gakkō=ni] omot-ta / kanzi-ta*. '*I found/felt [it school].'

Given that in the small-clause complements in (41) and (42), the theme θ -role can only be discharged by the NA stem *yukai-ni* and the VA stem *tanosi-ku* onto their argument *sore*, these stems must be lexical predicates. Thus, since the NA stem *yukai-ni*, as a lexical predicate, must discharge its theme θ -role onto an argument, the nominal-adjectival construction *sore=wa*

yukai-de ar-u in (33) can be analysed underlyingly as illustrated in (44), where the higher NP *sore_i* controls the *PRO_i* with the same index:

(44) [*sore_i=wa* [*PRO_i yukai-ni-te*] *t_i ar-ru-Top⁰∅*]

which consists of the syntactically obligatory construction [*sore_i=wa t_i ar-ru-Top⁰∅*] 'it exists' combined with the syntactically adjunctive construction [*PRO_i yukai-ni-te*] 'lit. (with) being pleasant', which surfaces as [*PRO_i yukai-de*]. If the syntactic structure of *sore=wa yukai-de aru* were [*sore_i=wa* [*yukai-ni-te*] *t_i ar-ru-Top⁰∅*] without PRO, the predicative NA *yukai-ni* would lack an argument onto which it could discharge its theme θ -role, whereas the predicative verb *ar-* would have its theme argument *sore*. Furthermore, given that the ConjP [*PRO_i yukai-ni-te*] constitutes atemporal domain, as discussed in Section 2, PRO cannot receive Case. The postulation of PRO in the nominal-adjectival construction is, therefore, not an analytical choice but a theoretical necessity, as PRO is the only category that requires a θ -role while being unable to receive Case.

Why, then, must the nominal-adjective phrase (NAP) [*PRO yukai-ni*] constitute the complement of the adjunctive conjunction *-te* and, in the form of the ConjP [*PRO yukai-ni-te*], be adjoined as an adjunct to the autonomous existential construction [*sore_i=wa t_i aru Top⁰∅*]? This is because **the NA stem *yukai-ni*, unlike the VA stem *tanosi-ku*, cannot function as a predicate in its bare *-ni* form outside small clauses that serve as complements of cognition or perception verbs**, as in (40), which is demonstrated in (44) – (47):

(45) *Sore=wa tanosi-ku, yukai-de aru.* 'It enjoyable, is pleasant.'

(46) **Sore=wa tanosi-ku, yukai-ni.* 'It enjoyable, pleasant.'

(47) *Sore=wa yukai-de ari, tanosi-i.* 'It is pleasant, enjoyable.'

(48) **Sore-wa yukai-ni, tanosi-i.* 'It pleasant, enjoyable.'

Whereas the VA stem derived from the VA radical *tanosi-* can function as a predicate in the form *tanosi-ku*, with the VA thematic suffix allomorph *-ku*, at the end of non-sentence-final coordinated clauses as in (44), and in the form *tanosi-i*, with the VA thematic suffix allomorph *-i*, at the end of sentence-final clauses as in (46), **the NA stem *yukai-ni* derived from the NA**

radical *yukai*- cannot function as a predicate in its bare form either at the end of non-sentence-final coordinated clauses or at the end of sentence-final clauses, as indicated in (45) and (47).

For this reason, it must constitute the complement of the adjunctive conjunction *-te* and, in the form of the ConjP [*PRO yukai-ni-te*] — surfacing as [*PRO yukai-de*] — be adjoined as an adjunct to the one-argument existential construction headed by the verb *ar-*. Only in this way — as it were, by 'parasitising' on the existential construction — can it function predicatively outside small clauses that serve as complements of cognition or perception verbs. This is why the NA construction [*sore_i=wa [PRO_i yukai-de] t_i ar-u-Top⁰∅*] — underlyingly, [*sore_i=wa [PRO_i yukai-ni-te] t_i ar-ru-Top⁰∅*] — appears superficially identical to the so-called copular construction [*sore_i=wa [gakkō=de] t_i ar-u-Top⁰∅*] — underlyingly, [*sore_i=wa [gakkō=ni-te] t_i ar-ru-Top⁰∅*].

As already discussed, the NA stem is a lexical predicate that requires its argument(s), onto which it discharges its θ -role(s). By contrast, since the underlying PP [*gakkō=ni*] in the so-called copular construction in (32) does not constitute a lexical predicate, unlike the NA [*yukai-ni*], it does not require any argument, and therefore the PP [*gakkō=ni*] does not contain PRO. Accordingly, the semantic core constructions in (40) and (44), as well as their schematic expressions, are underlyingly different, as shown in (49) and (50):

$$(49) \text{Sore=wa gakkō=de ar-u:} \quad [\text{VP}_1 [\text{ConjP} [\text{PP gakkō=ni}] \text{-te}] [\text{VP}_0 \text{sore}_i \text{ar-}]] \\ \text{EXIST(sore)} \wedge \text{IN(rEXIST(sore), gakkō)}$$

$$(50) \text{Sore=wa yukai-de ar-u:} \quad [\text{VP}_1 [\text{ConjP} [\text{NAP PRO}_i \text{yukai-ni}] \text{-te}] [\text{VP}_0 \text{sore}_i \text{ar-}]] \\ \text{EXIST(sore)} \wedge \text{PLEASANT(sore)}$$

That said, since both constructions are schematically representable as conjunctive propositions, they exhibit the same pattern of partial negation, in which only the additional proposition following $\wedge$ is negated, as shown in (51) and (52):

$$(51) \text{Sore=wa gakkō=de=wa (*ara-)nai.} \\ \text{EXIST(sore)} \wedge \neg \text{IN(rEXIST(sore), gakkō)}$$

$$(52) \text{Sore=wa yukai-de=wa (*ara-)nai.} \\ \text{EXIST(sore)} \wedge \neg \text{PLEASANT(sore)}$$

Furthermore, there also exist NA stems such as *suki-ni* 'to like' and *kirai-ni* 'to dislike' which function as **two-argument predicates**. The constructions containing such NA stems, however, share the same basic structure as those containing one-argument NA stems such as *sizuka-ni* and *yukai-ni*: a NAP containing PRO is adjoined as an adjunct, in the form of a ConjP headed by the adjunctive conjunction *-te*, to the VP headed by the existential-verb stem *ar-*, as shown in (53):

- (53) *Kare=wa kanozyo=ga suki=de aru.* 'He likes her.'
 $[_{VP_i} [_{ConjP} [_{NAP} PRO_i \text{ kanozyo suki-ni}] -te] [_{VP_o} \text{ kare}_i \text{ ar-}]]$
 $EXIST(kare) \wedge LIKE(kare, kanozyo)$

Its default negation pattern is likewise one of partial negation, as in (54):

- (54) *Kare=wa kanozyo=ga suki=de=wa (*ara-)nai.*
 $EXIST(kare) \wedge \neg LIKE(kare, kanozyo)$

By contrast, the default negation pattern of the VA construction in (55) is the total negation in (56), not the partial negation in (57):

- (55) *Sore=wa tanosi-i:* ENJOYABLE(sore)
(56) *Sore=wa tanosi-ku nai:* $\neg$ ENJOYABLE(sore)
(57) *Sore=wa tanosi-ku=wa nai:* $\neg$ ENJOYABLE(sore) $\wedge$ SOME-OTHER-STATE(sore)

Accordingly, the VA construction *sore=wa tanosi-i* in (55) is fundamentally different from the NA construction *sore=wa yukai-de aru*, in that the former is schematically representable as the atomic proposition ENJOYABLE(sore), as in (55), whereas the latter can be represented as the conjunctive proposition $EXIST(sore) \wedge PLEASANT(sore)$, as in (50). It follows that the stem allomorph *at-* of the verb *ar-u* occurring in the past form *tanosi-k-at-ta*, for example, does not instantiate the one-argument existential verb *ar-* 'to exist' used in both so-called copular and nominal-adjectival sentences. Rather, it is an allomorph of the light verb stem *ar-* inserted as a matter of formal necessity in order to concatenate the verbal past-tense suffix *-ta* with the non-verbal VA stem *tanosi-ku*, as in $\text{tanosi-ku} + -ta \rightarrow \text{tanosi-ku} + \text{ar-} + -ta \rightarrow \text{tanosi-k-at-ta}$.

Thus, the syntactic derivation of the so-called copular, nominal-adjectival and verbal-adjectival

constructions can be represented in (58)–(61):

(58) *Sore=wa gakkō=de aru.* '(lit.) It exists, with its existence located in the domain of "school".'

Adjunct	Core construction
[=ni]	[ar-]
→ [gakkō=ni]	→ [sore ar-]
→ [[gakkō=ni] -te]	
	→ [[[gakkō=ni] -te] [sore ar-]]
	→ [[[[gakkō=ni] -te] [sore ar-]] -ru]
	→ [[[[[gakkō=ni] -te] [sore ar-]] -ru] Top ⁰ ∅]
	→ [sore _i =wa [[[[[gakkō=ni] -te] [t _i ar-]] -ru] Top ⁰ ∅]]

(59) *Sore=wa yukai-de aru.* '(lit.) It, being pleasant, exists.'

Adjunct	Core construction
[yukai-ni]	[ar-]
→ [PRO yukai-ni]	→ [sore ar-]
→ [[PRO yukai-ni] -te]	
	→ [[[PRO _i yukai-ni] -te] [sore _i ar-]]
	→ [[[[PRO _i yukai-ni] -te] [sore _i ar-]] -ru]
	→ [[[[[PRO _i yukai-ni] -te] [sore _i ar-]] -ru] Top ⁰ ∅]
	→ [sore _i =wa [[[[[PRO _i yukai-ni] -te] [t _i ar-]] -ru] Top ⁰ ∅]]

(60) *Sore=wa tanosi-i.* 'It is enjoyable.'

Core construction
[tanosi-ku]
→ [sore tanosi-ku]
→ [[sore tanosi-ku] T ⁰ ∅]
→ [[[sore tanosi-ku] T ⁰ ∅] Top ⁰ ∅]
→ [sore _i =wa [[[[t _i tanosi-ku] T ⁰ ∅] Top ⁰ ∅]]

In the VA non-past form in (60), the sequence $[-ku + T^0\emptyset]$ is morphophonologically realised as $-i$, yielding the surface form *tanosi-i* from the underlying elsewhere form *tanosi-ku*. The exponent $-i$ may therefore be analysed either as a contextual allomorph $-i$ of the VA thematic suffix $-ku$ in the non-past environment or, alternatively, as an amalgamated realisation of the sequence $[-ku + T^0\emptyset]$, as stated in Makino (2005¹: 14).

(61) *Sore=wa tanosi-k-at-ta*. 'It was enjoyable.'

Core construction

$[tanosi-ku]$

$\rightarrow [sore\ tanosi-ku]$

$\rightarrow [[sore\ tanosi-ku]\ -ta]$

$\rightarrow [[sore\ tanosi-ku]\ ar-\ -ta]$

$\rightarrow [[[sore\ tanosi-ku]\ ar-\ -ta]\ Top^0\emptyset]$

$\rightarrow [sore_i=wa\ [[[t_i\ tanosi-ku]\ ar-\ -ta]\ Top^0\emptyset]]$

In the analysis of the VA past form in (61), the choice between the non-past $T^0\emptyset$, realised as -0 , and the past suffix $-ta$ is treated as a paradigmatic selection made at a single derivational stage, since the $[-past]\ -0$ and the $[+past]\ -ta$ constitute a paradigm whose terms acquire linguistic value solely through their opposition at the same point of speech. When the past suffix $-ta$ is selected, it proves incompatible with non-verbal predicates such as *tanosi-ku*, thereby triggering the insertion of the light verb stem *ar-* as a repair strategy.

Returning to the NA *sizukada*, the morphosyntactic structure of the non-past form *sizuka-de#ar-u* may be represented, under the above analysis, as in (62), where '#' indicates a morphological word boundary that coincides with a prosodic word boundary, and 'them.suff.', 'conj.' and 'inflex.suff.' abbreviate 'thematic suffix', 'conjunction' and 'inflexional suffix', respectively:

(62)

Word (when <i>-ni</i> , <i>-ni-te</i> → <i>-n-</i> , <i>-d-</i>)					Surface form
Word			Word		
NA stem		Ending	Existential-verb stem	Ending	
NA radical	NA them.suff.	Conj.	Existential-verb radical	Inflex.suff.	
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-ru</i> [–past]	
<i>sizuka-de#ar-u</i>					

whereas the morphosyntactic structure of the verbal-adjective forms *taka-i* 'it is high/tall' and

taka-k-at-ta 'it was high/tall' can be illustrated as in (63):

(63)

Word				Surface form
VA Stem: sentential			Ending	
VA Stem: lexical		Light-verb stem		
VA radical	VA thematic suffix	Light-verb radical	Inflexional suffix	
<i>taka-</i>	<i>-i</i>	—	<i>-0 [-past]</i>	
<i>taka-</i>	<i>-ku</i>	<i>ar-</i>	<i>-ta [+past]</i>	

<i>taka-i- -0</i>
<i>taka-k-at-ta</i>

In *sizuka-de#ar-u*, it is possible to insert at the sentential level the focus particle =*mo* 'too, also' or the topicalisation/contrast particle =*wa* 'as for, etc.' — both of which are enclitics — immediately after *sizuka-de*, without insertion of new constituents or deletion of existing ones, as shown in (64):

(64) *sizuka-de#ar-u* → *sizuka-de=mo#ar-u* / *sizuka-de=wa#ar-u*

In *sizuka-de#ar-u*, therefore, both *sizuka-de* and *ar-u* constitute morphologically autonomous words.

5. The contracted forms *sizuka-d-a* / *sizuka-d-at-ta* from *sizuka-de#ar-u* / *sizuka-de#at-ta* and other forms

The surface non-past form of the NA *sizukada* shows free variation between *sizuka-de#ar-u* and *sizuka-d-a*, the former being more formal and literal. It can be observed that, in the latter, the vowel *-e* in the suffix sequence *-de* — underlyingly *-ni-te* — is deleted before another vowel *a-*. Furthermore, neither the final consonant *-r* of the existential verb stem *ar-* nor the allomorph *-u* of the non-past suffix *-ru* appear. The more informal and colloquial form *sizuka-d-a* nevertheless has the same non-past value as *sizuka-de#ar-u*. From this, it can be assumed that, in *sizuka-d-a*, the non-past suffix does not bear the overt form *-ru* but the phonologically null *-0*, just as it does in VA non-past forms such as *taka-i-0*, and that the final *-r* of the existential verb stem is deleted at the surface level by virtue of its position at the word-final boundary. Thus, the derivation from the underlying form (UF) to the surface form (SF) via the hypothesised intermediate form (IF) can be assumed as in (65):

(65) UF: *sizuka-ni-te#ar-0* → IF: **sizuka-de#a* → SF: *sizuka-d-a*

Historically, although a so-called copula, the form =*de#a*, which yielded the modern =*d-a*, can be attested in Azuchi–Momoyama-period materials: *bonnin yorimo giūzaini fuxōzuru cotodea* '(it) is a reason for imposing a heavier guilt (on him) than on an ordinary person' (Amakusa Edition of *Esopo no fabulas*, 1593, p.475, ll.10-11, in Text Data Sets for Research on the History of Japanese).

Taking as an example the reduction in *kanozyo=wa sizuka-de ar-u* → *kanozyo=wa sizuka-d-a*, the reduction of *ar-u* to *a* can be accounted for in pragmatic and informational terms. First, as already discussed in Section 4, the core predicate in this construction is informationally not the verb *ar-* but the NA stem *sizuka-ni* contained within the surface NA stem–conjunction complex *sizuka-de* — underlyingly *sizuka-ni-te*. Second, since the topicalised phrase *kanozyo=wa* generally presupposes, at the discourse level, the existence of *kanozyo*, and since the occurrence of the constituent *sizuka-de* entails the existence of *kanozyo* by virtue of the conditional $\text{QUIET}(\text{kanozyo}) \Rightarrow \text{EXIST}(\text{kanozyo})$, given that *kanozyo* is treated as an entity in $\text{QUIET}(\text{kanozyo})$, the verb *ar-* becomes informationally redundant when it once again asserts *kanozyo*'s existence. Third, in a sentence in which $\text{QUIET}(\text{kanozyo})$ is presented as new information, $\text{EXIST}(\text{kanozyo})$ is presupposed and thus treated as given, owing to the same conditional $\text{QUIET}(\text{kanozyo}) \Rightarrow \text{EXIST}(\text{kanozyo})$. In this context, $\text{EXIST}(\text{kanozyo})$ is pragmatically and informationally less significant than the focus component $\text{QUIET}(\text{kanozyo})$. It may therefore be assumed that **this low pragmatic and informational salience is reflected in the phonological reduction of *ar-u* to *a*.**

The reduction of the underlying *-ni-te ar-0* to the surface *-d-a*, via *-de ar-0*, on the other hand, can be accounted for by the fixed order of these concatenated morphemes and its frequency. In existence-asserting and state-specifying NA constructions, the NA stem in *-ni*, the conjunction *-te* and the verbal stem *ar-* — which as a whole corresponds to the schematic expression $\text{EXIST}(\) \wedge \text{NA-STATE}(\)$ and in which the NA stem in *-ni*, *-te* and *ar-* represent $\text{NA-STATE}(\)$, $\wedge$ and $\text{EXIST}(\)$, respectively — is invariably ordered in this way. **The very high frequency of this fixed order in actual utterances is assumed to have led to the phonological reduction of *-ni-te ar-0* to *-d-a*, via *-de ar-0*,** just as in the contractions *si-te=i-ru* → *si-te-ru* 'to be doing something' and *si-te=sima-u* → *si-t-ya-u* 'to end up doing sth.' or 'to do sth. inadvertently or unintentionally' in colloquial Japanese. The reduction of the underlying *-ni-te ar-ta* to the surface *-d-at-ta*, via *-de at-ta*, in the past form — as in *kanozyo=wa sizuka-de at-ta* → *kanozyo=wa sizuka-d-at-ta* — can be likewise explained.

Under this analysis, the morphological structure of *sizuka-d-a* and *sizuka-d-at-ta*, as well as their non-contracted forms *sizuka-de#ar-u* and *sizuka-de#at-ta*, can be represented as in (66):

(66)

Word (when <i>-ni-te</i> → <i>-d-</i>)					Surface form
Word			Word		
Stem		Ending	Existential-verb stem	Ending	
Radical	Them.suff.	Conj.	Existential-verb radical	Inflex. suff.	
<i>sizika-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-0</i> [–past]	
<i>sizika-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-ru</i> [–past]	
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-tar</i> [+past]	
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-tar</i> [+past]	

<i>sizuka-d-a</i>
<i>sizuka-de#ar-u</i>
<i>sizuka-de#at-ta</i>
<i>sizuka-d-at-ta</i>

The attributive form — the so-called *rentaikei* — exhibits practically the same type of free variation between *sizuka-n-ar-u* and *sizuka-n-a* as between *sizuka-de#ar-u* and *sizuka-d-a* in (66), the former being much more formal and literal. In these forms, *sizuka-n-* can be interpreted as the underlying NA stem *sizuka-ni* deleting its final vowel *-i* before the following vowel *a-*, **without the intervention of the adjunctive conjunction *-te*, unlike other forms involving *-te***. Note that, historically as well, the etymon of *-n-ar-u* was in fact an older form *-ni#ar-u* without *-te*. Thus, the derivation of each form from the UF to the SF can be assumed as in (67) and (68), respectively:

(67) UF: *sizuka-ni#ar-ru* → SF: *sizuka-n-ar-u*

(68) UF: *sizuka-ni#ar-0* → SF: *sizuka-n-a*

The morphosyntactic structure of the forms in (67) and (68) — including the imperative form *sizuka-n-ar-e* 'Be quiet', which is much more formal and literal than the current *sizuka-de#ar-e*, and the everyday conditional form *sizuka-n-ar-a(-ba)* 'if it is quiet', with a semantically synonymous yet morphologically distinct form *sizuka-de#ar-eba* — is represented in (69):

(69)

Word (when <i>-ni</i> , <i>-ni-te</i> → <i>-n-</i> , <i>-d-</i>)					Surface form
Word			Word		
Stem		Ending	EV stem	Ending	
Radical	Them.suff.	Conjunct.	EV radical	Inflex.suff.	
<i>sizuka-</i>	<i>-ni</i>		<i>ar-</i>	<i>-0</i> [–past]	
<i>sizuka-</i>	<i>-ni</i>		<i>ar-</i>	<i>-ru</i> [–past]	
<i>sizuka-</i>	<i>-ni</i>		<i>ar-</i>	<i>-e</i>	
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-e</i>	

<i>sizuka-n-a</i>
<i>sizuka-n-ar-u</i>
<i>sizuka-n-ar-e</i>
<i>sizuka-de#ar-e</i>

<i>sizuka-</i>	<i>-ni</i>		<i>ar-</i>	<i>-a(-ba)</i>	<i>sizuka-n-ar-a(-ba)</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-reba</i>	<i>sizuka-de#ar-eba</i>

Other forms containing the existential-verb stem *ar-* with verbal endings *-te*, *-temo*, *-tatte*, *-tar-ō*, *-tar-a(-ba)*, *-tar-i*, *-ru*, *-reba*, *-jō*, *-e*, *-tutu* and *-nagara* can be analysed as shown in (70):

(70)

Word (when <i>-ni-te</i> → <i>-d-</i>)				
Word			Word	
Stem		Ending	EV stem	Ending
Radical	Them.suff.	Conjunct.	EV radical	Inflex.suff.
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-te</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-temo</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-tatte</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-jō</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-jō</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-tar-ō</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-tar-ō</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-tar-a(-ba)</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-tar-a(-ba)</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-tar-i</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-tar-i</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-tutu</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-nagara</i>

Surface form
<i>sizuka-de#at-te</i>
<i>sizuka-de#at-temo</i>
<i>sizuka-de#at-tatte</i>
<i>sizuka-d-ar-ō</i>
<i>sizuka-de#ar-ō</i>
<i>sizuka-d-at-tar-ō</i>
<i>sizuka-de#at-tar-ō</i>
<i>sizuka-d-at-tar-a(-ba)</i>
<i>sizuka-de#at-tar-a(-ba)</i>
<i>sizuka-d-at-tar-i</i>
<i>sizuka-de#at-tar-i</i>
<i>sizuka-de#ari-tutu</i>
<i>sizuka-de#ari-nagara</i>

Finally, in the underlying forms of *sizuka-de* and *sizuka-demo* — namely, *sizuka-ni-te* and *sizuka-ni-temo* — it is possible to insert the enclitic particles =*wa* or =*mo* at the sentential level between the NA stem *sizuka-ni* and the conjunctions *-te* and *-temo*, as shown in (71) and (72):

(71) *sizuka-ni* + =*wa* + *-te* → *sizuka-ni-te*=*wa#ar-te* → *sizuka-de*=*wa#at-te*

(72) *sizuka-ni* + =*wa* + *-temo* → *sizuka-ni-te*=*wa#ar-temo* → *sizuka-de*=*wa#at-temo*

In this operation, through the insertion of =*wa*, the conjunctive suffixes *-te* and *-temo* are separated from the stem *sizuka-ni* and thereby rendered stemless, whereupon the stem *ar-* of the existential verb *ar-u* is inserted before them, resolving this stemlessness. At the same time, the adjunctive conjunction *-te* is supplied after the VA stem *sizuka-ni*, and the resultant ConjP *sizuka-ni-te* serve as an adjunct to the VP headed by the verbal stem *ar-*, as stated in Section 4. This operation can also be applied to the underlying form of *sizuka-datte* — namely, *sizuka-ni-tatte*, hypothesised in (31) — as indicated in (73):

(73) *sizuka-ni* + =wa + *-tatte* → *sizuka-ni-te*=wa#*ar-tatte* → *sizuka-de*=wa#*at-tatte*

We are inclined to posit *sizuka-ni-tatte* as the UF of *sizuka-datte* — a form that is neither diachronically attested nor consistent with native-speaker intuition — solely on the basis of the homogeneity of the processes illustrated in (71) – (73).

6. Other formal differences between the so-called copular constructions and the nominal-adjectival constructions

As discussed in Section 4, the so-called copular sentences and the nominal-adjectival sentences are underlyingly distinct in structure, in that the latter crucially make recourse to a PRO construction. In addition to this difference, there are further notable oppositions between the two constructions, as illustrated in (74) and (75).

(74) In contemporary standard Japanese, **NA radicals followed by *-d-a*** — for example, *sizuka-d-a*, underlyingly *sizuka-ni-te ar-0* — **exhibit a range of inflected forms that contain the thematic suffix *-ni* but not the conjunction *-te***. These include *sizuka-ni* 'quiet; quietly', *sizuka-n-a daiti* 'a land that is quiet', *sizuka-n-ar-u daiti* 'idem.' (more formal and literal), *sizuka-n-ar-e* 'be quiet' (highly formal and literal), *sizuka-n-ar-a(-ba)* 'if it is quiet' and *sizuka-n-a no(=wa/=ga/=o/=ni/=de/=to)* '(the fact) that it is quiet'.

By contrast, **NPs followed by *=d-a*** — for example, *gakusei=d-a*, underlyingly *gakusei=ni-te ar-0* — **do not generally exhibit inflected forms with *-ni* but without *-te***. First, when the underlying form *gakusei=ni-te ar-0* does not involve the existential verb *ar-0*, it invariably surfaces with the conjunction *-te*, yielding *gakusei=ni-te*, which in turn surfaces as *gakusei=de*: it never appears in the form *gakusei=ni*. Second, forms such as **gakusei=n+a wakamono* 'a youth that is a student', **gakusei=n+ar-u wakamono* 'idem.' and **gakusei=n+ar-e* 'be a student' — whose standard-language counterparts would be *gakusei=no wakamono*, *gakusei=de#ar-u wakamono* and *gakusei=de#ar-e*, respectively — are not normally used in contemporary Japanese.

The only exceptions are, to our knowledge, *gakusei=n-a no(=wa/=ga/=o/=ni/=de/=to)* '(the fact) that it is a student' and *gakusei=n-ar-a(-ba)* 'if it is a student', in which the NP *gakusei* refers to an individual. The first nominalised expression *=n-a no* is assumed to derive from the

underlying sequence =*ni* *ar-0* *no*, where *no* functions as a formal noun or nominaliser. The second conditional expression =*n-ar-a(-ba)*, derived from the underlying =*ni* *ar-a(-ba)*, appears to be a fossilised form inherited from classical Japanese, where =*n-ari*, derived from the underlying =*ni* *ar-0*, functioned as the standard so-called copula. However, the form =*nara(ba)* can nowadays occur immediately after non-nominal expressions, as in *Karera=ga koko=ni kuru=nara(-ba)* 'if they come here', which indicates that =*nara(-ba)* already functions as a conjunction.

(75) In certain fixed expressions such as *NPs=n-a mono=de* and *NPs=n-ar-u mono* — for instance, *watasi=wa gakusei=n-a mono=de* 'because/since I am (qualified/characterised as) a student' and *gakusei=n-ar-u mono* 'that which is (qualified/characterised as) a student' — as well as in colloquial expressions such as *gakusei=n-a kanzi*, which is nearly synonymous with *gakuseiteki-n-a kanzi* 'a student-like impression', inflected forms with *-ni* but without *-te* can be used. In these examples, nevertheless, **the NP *gakusei* does not seem to refer to an individual but rather to properties, qualities or characteristics, and thus is semantically closer to NA radicals.**

7. Analysis summary — Conclusion

(i) The inflected forms of the NA *sizukada* can be morphologically analysed as shown in (76):

(76)

Word (when <i>-ni</i> , <i>-ni-te</i> → <i>-n-</i> , <i>-d-</i>)				
Word			Word	
Stem		Ending	EV stem	Ending
Radical	Them.suff.	Conjunct.	EV radical	Inflex.suff.
<i>sizuka-</i>	<i>-ni</i>			
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>		
<i>sizuka-</i>	<i>-ni</i>	<i>-temo</i>		
<i>sizuka-</i>	<i>-ni</i>	<i>-tatte</i>		
<i>sizuka-</i>	<i>-ni</i>		<i>ar-</i>	<i>-0</i> [–past]
<i>sizuka-</i>	<i>-ni</i>		<i>ar-</i>	<i>-ru</i> [–past]
<i>sizuka-</i>	<i>-ni</i>		<i>ar-</i>	<i>-e</i>
<i>sizuka-</i>	<i>-ni</i>		<i>ar-</i>	<i>-a(-ba)</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-0</i> [–past]
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-ru</i> [–past]
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-e</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-reba</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-jō</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-jō</i>

Surface form
<i>sizuka-ni</i>
<i>sizuka-de</i>
<i>sizuka-demo</i>
<i>sizuka-datte</i>
<i>sizuka-n-a</i>
<i>sizuka-n-ar-u</i>
<i>sizuka-n-ar-e</i>
<i>sizuka-n-ar-a(-ba)</i>
<i>sizuka-d-a</i>
<i>sizuka-de#ar-u</i>
<i>sizuka-de#ar-e</i>
<i>sizuka-de#ar-eba</i>
<i>sizuka-d-ar-ō</i>
<i>sizuka-de#ar-ō</i>

<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-te</i>	<i>sizuka-de#at-te</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-temo</i>	<i>sizuka-de#at-temo</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-tatte</i>	<i>sizuka-de#at-tatte</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-tar [+past]</i>	<i>sizuka-d-at-ta</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-tar [+past]</i>	<i>sizuka-de#at-ta</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-tar-ō</i>	<i>sizuka-d-at-tar-ō</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-tar-ō</i>	<i>sizuka-de#at-tar-ō</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-tar-a(-ba)</i>	<i>sizuka-d-at-tar-a(-ba)</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-tar-a(-ba)</i>	<i>sizuka-de#at-tar-a(-ba)</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-tar-i</i>	<i>sizuka-d-at-tar-i</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-tar-i</i>	<i>sizuka-de#at-tar-i</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-tutu</i>	<i>sizuka-de#ari-tutu</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-nagara</i>	<i>sizuka-de#ari-nagara</i>

(ii) The NA radical is the element that encodes the lexical meaning specific to each nominal adjective. It can occur independently as a constituent of derived or compound words — for example, *hukisoku-* in *hukisoku-sa* 'irregularity' and *hukisoku-dōsi* 'irregular verb' — but it cannot, on its own, be used as a syntactic constituent belonging to the category VA at the sentential level — e.g. **Kare=wa sizuka mat-ta* 'he danced quietly', **sizuka nar-u / *sizuka su-ru* 'to become/make quiet/silent' or **sizuka-a hito* 'a quiet person'. Compared to VA radicals, however, NA radicals exhibit greater independence: in more colloquial speech, they may occur as predicates of main clauses without *-d-a* or *=de aru*, as in *kare(=wa) sugoku **sizuka*** 'he very quiet' vs. the more canonical *kare=wa sugoku **sizuka-d-a** / **sizuka=de aru*** 'he is very quiet'.

(iii) The function of the NA thematic suffix *-ni* is, on the other hand, to license the NA radical as a syntactic constituent of category NA within sentential phrase structure. **It is precisely this *-ni* which constitutes the defining constituent of NAs** — an indispensable morphosyntactic and semantic constituent for the construction of the NA stem — and, as noted in (ii), NA radicals such as *sizuka-* on their own thus cannot be used as NAs in a sentence. Only when combined with *-ni* do they become syntactically eligible to function as constituents of category NA — for example, *kare=wa sizuka-ni mat-ta* 'he danced quietly', *sizuka-ni nar-u / sizuka-ni su-ru* 'to become/make quiet', or *sizuka-n-a hito* 'a quiet person'. The NA thematic suffix *-ni* functions morphosyntactically differently from the formally identical postposition *=ni*, yet shares the same semantics of 'in/at', which appears to be a semantically indispensable element for NAs.

(iv) When the thematic suffix *-ni* is followed by the inflexional suffixes *-te*, *-temo*, *-tatte*, it undergoes amalgamation, yielding *-de*, *-demo*, *-datte*. Furthermore, the thematic suffix *-ni* and the

contracted form *-de* of the sequence *-ni-te* often deletes its final *-e* before the allomorphs of the stem *ar-* of the existential verb *ar-u*. Where this deletion is optional in many cases, the non-deletion form is more formal and literal. It should be noted that, although the derivation *-ni-tatte* → *-datte* is neither diachronically attested nor consistent with our native-speaker intuition, we nevertheless posit, for example, *sizuka-ni-tatte* as the underlying form of *sizuka-datte* on the basis of the homogeneity of the following processes:

sizuka-ni + *=wa* + *-te* → *sizuka-ni-te=wa#ar-te* → *sizuka-de=wa#at-te*

sizuka-ni + *=wa* + *-temo* → *sizuka-ni-te=wa#ar-temo* → *sizuka-de=wa#at-temo*

sizuka-ni + *=wa* + *-tatte* → *sizuka-ni-te=wa#ar-tatte* → *sizuka-de=wa#at-tatte*

(v) In sum, the core of NAs lies in the **NA stem in *-ni***, as in *sizuka-ni* 'to be quiet/silent' and *suki-ni* 'to like'. These NA stems are lexical predicates that require arguments — the argument structure [theme] inherent in *sizuka-ni*, and [experiencer, theme] in *suki-ni*. However, outside small clauses functioning as complements of cognition or perception verbs, as in *Watasi=wa [sore=o yukai-ni] omot-ta / kanzi-ta* 'I found / felt [it pleasant]', such NA stems cannot function as predicates in their bare *-ni* form.

Consequently, an NA stem forms a nominal-adjective phrase accompanied by PRO, as in [*PRO sizuka-ni*], which serves as the complement of the adjunctive conjunction *-te*, yielding a ConjP [*PRO sizuka-ni-te*]. This ConjP is then adjoined to the VP headed by the existential-verb stem *ar-* within a one-argument construction, as in [*kanozyo_i=wa t_i ar-ru-Top⁰∅*] 'she exists'. Only in this configuration can the NA stem function as the predicate of an independent sentence, as in [*kanozyo_i=wa [PRO_i sizuka-ni-te] t_i ar-ru-Top⁰∅*], which surfaces as [*kanozyo_i=wa [PRO_i sizuka-de] t_i ar-u-Top⁰∅*] 'she is quiet' (literally, 'she, being quiet, quiet'), schematically representable as *EXIST(she) ∧ QUIET(she)*, whose default negation pattern is partial, namely *EXIST(she) ∧ ¬QUIET(she)*.

(vi) By contrast, the so-called copular construction [*kanozyo_i=wa [gakusei=de] t_i ar-u-Top⁰∅*] — more underlyingly [*kanozyo_i=wa [gakusei=ni-te] t_i ar-ru-Top⁰∅*] — for example, consists syntactically of the one-argument core construction [*kanozyo_i=wa t_i ar-ru-Top⁰∅*] 'she exists' and the adjunctive construction [*gakusei=ni-te*] 'and (her existence is located) in the conceptual domain of "student"' or, more practically, 'and (it is) as a student', which surfaces as [*gakusei-de*].

Semantically, this construction is both existence-asserting and existence-domain-specifying. The VP [*kanozyo ar-*] in the syntactically obligatory construction [*kanozyo_i=wa t_i ar-ru-Top⁰∅*] asserts her existence. On the other hand, the underlying postposition *=ni* in the syntactically adjunctive ConjP [*gakusei=ni-te*] behaves as if it were a two-argument predicate, and locates her existence denoted by the VP [*kanozyo ar-*] within the conceptual domain of 'student' referred to by the NP [*gakusei*].

Schematically, this construction can be represented as $\text{EXIST}(\text{she}) \wedge \text{IN}(\text{rEXIST}(\text{she}), \text{student})$, whose default negation pattern is partial, namely $\text{EXIST}(\text{she}) \wedge \neg \text{IN}(\text{rEXIST}(\text{she}), \text{student})$. Here the function *r* in $\text{rEXIST}(\text{she})$ — abbreviated from 'reified' — maps the proposition $\text{EXIST}(\text{she})$ 'she exists' to the corresponding reified eventuality $\text{rEXIST}(\text{she})$, that is, an eventive entity representing 'her existence'. Since the PP [*gakusei=ni*] does not constitute a lexical predicate, it does not require any argument, and therefore the adjunctive construction [*gakusei=ni*] does not contain PRO, unlike the nominal-adjectival constructions.

(vii) Whereas the so-called copular sentences and the nominal-adjectival sentences consist syntactically of an existence-asserting, one-argument obligatory construction [*NP₁=wa t_i ar-Top⁰∅*] combined with an existence-domain-specifying adjunctive ConjP [*NP₂=ni-te*] or a state-specifying adjunctive ConjP [*PRO_i NA radical-ni-te*], respectively, and are schematically representable as conjunctive $\text{EXIST}(\text{NP}_1) \wedge \text{IN}(\text{rEXIST}(\text{NP}_1), \text{NP}_2)$ or $\text{EXIST}(\text{NP}_1) \wedge \text{NA-STATE}(\text{NP}_1)$, respectively — whose default negation pattern is partial, namely $\text{EXIST}(\text{NP}_1) \wedge \neg \text{IN}(\text{rEXIST}(\text{NP}_1), \text{NP}_2)$ and $\text{EXIST}(\text{NP}_1) \wedge \neg \text{NA-STATE}(\text{NP}_1)$ — the verbal-adjectival sentences consist only of a state-specifying obligatory construction [*NP₁=wa t_i VA stem-Top⁰∅*] without any adjunct and can be represented as atomic VA-STATE (NP) — whose default negation pattern is total, namely $\neg \text{VA-STATE}(\text{NP})$.

(viii) In certain fixed expressions such as *NP=n-a mono=de* and *NP=n-ar-u mono*, and in colloquial expressions such as *NP=n-a kanzi*, inflected forms with *=ni* but without *-te* of the so-called copula *=d-a* can be used. In these expressions, nevertheless, the NP does not seem to refer to an individual but rather to properties, qualities or characteristics, and thus is semantically closer to NA radicals. Accordingly, it is also possible to interpret **the NP=n-** in those expressions as **constituting, at the syntactic level** rather than at the lexical level, **the NA stem NP-ni, surfacing as NP-n-**.

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